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Separate Mechanisms Recruited by Exogenous and Endogenous Spatial Cues: Evidence From a Spatial Stroop Paradigm

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The present experiments tested whether endogenous and exogenous cues produce separate effects on target processing. In Experiment 1, participants discriminated whether an arrow presented left or right of fixation pointed to the left or right. For 1 group, the arrow was preceded by a peripheral noninformative cue. For the other group, the arrow was preceded by a central, symbolic, informative cue. The 2 types of cues modulated the spatial Stroop effect in opposite ways, with endogenous cues producing larger spatial Stroop effects for valid trials and exogenous cues producing smaller spatial Stroop effects for valid trials. In Experiments 2A and 2B, the influence of peripheral noninformative and peripheral informative cues on the spatial Stroop effect was directly compared. The spatial Stroop effect was smaller for valid than for invalid trials for both types of cues. These results point to a distinction between the influence of central and peripheral attentional cues on performance and are not consistent with a unitary view of endogenous and exogenous attention.

Keywords: endogenous attention, exogenous attention, spatial Stroop

Covert visual orienting allows high-ordered animals to select potentially relevant visual information from the environment without the need to move their eyes. The cognitive processes and neural substrates underlying this ability are the subject of a great deal of empirical study (see Klein, 2004, for a recent review).

A widely used framework used to explain covert visual orienting was proposed by Posner and colleagues (Posner, 1980; Posner & Cohen, 1984). According to this framework, the attentional system operates independently from sensory receptors to orient toward a region of space where a potentially relevant visual stimulus is located. A consequence of a shift of attention toward a relevant region of space is an enhancement in processing stimuli inside the attended region, compared with stimuli located outside the attended region.

The spatial cuing paradigm introduced by Posner and colleagues has been used extensively to evaluate this framework (Posner, 1980; Posner & Cohen, 1984; Posner, Nissen, & Ogden, 1978). In this paradigm, subjects must detect, localize, or discriminate a target that appears either at a location previously indicated by a spatial cue or at some other location not indicated by a spatial cue. Typically, there is an improvement in performance for targets appearing at cued locations relative to targets at uncued locations, often referred to as a *facilitation effect*. This effect has been observed in a large number of tasks using behavioral measures (see Cave & Bishot, 1999, and Ruz & Lupiáñez, 2002, for recent reviews) but has also been observed electrophysiologically, with the amplitude of the P1 (80–130 ms) and N1 (150–200 ms) event-related-potential components over the posterior scalp being larger in the wave evoked by stimuli appearing at attended locations than in the wave evoked at unattended locations (e.g., Anllo-Vento, 1995; Di Russo, Martínez, & Hillyard, 2003; Eimer, 1993; Harter, Aine, & Shroeder, 1982; Mangun, 1995; Mangun & Hillyard, 1987).

The facilitation effect can be found both following central symbolic cues signaling the most likely target location (known as endogenous cues) and with peripheral, abrupt onset cues that do not predict the target location (known as exogenous cues). This result is consistent with the view that there are two modes by which attention can be oriented: a voluntary or *endogenous* mode that is responsive to internally developed expectancies and a reflexive or *exogenous* mode that is responsive to salient properties of external stimuli.

However, endogenous and exogenous cues are very different in some respects. Endogenous cues are typically informative, symbolic (i.e., they require interpretation), and presented centrally,

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whereas exogenous cues are typically noninformative, not symbolic, and presented peripherally. These differences have led to debate over whether the two modes of orienting triggered by exogenous and endogenous cues constitute two ways of “transporting” the very same attentional mechanism (the unitary view) or whether they recruit separate attentional systems. A variety of research strategies has been used to respond to this question, although no general agreement has yet been achieved.

One research strategy focuses on intrinsic properties of the orienting process per se, by analyzing possible differences in the speed and trajectory of orienting triggered by endogenous and exogenous cues. Early studies demonstrated that facilitation effects were smaller and appeared at shorter cue–target intervals with peripheral noninformative than with central informative cues (Jonides, 1981; Müller & Rabbitt, 1989). In addition, differences not only in size but in the nature of the effects have been found between purely exogenous and endogenous orienting. Posner and Cohen (1984) were the first to report a reversal of the facilitation effect at cue–target intervals (i.e., stimulus onset asynchronies [SOAs]) longer than 250 ms, an effect that occurs exclusively with peripheral noninformative cues. This negative cuing effect was termed inhibition of return (IOR; Posner, Rafal, Choate, & Vaughan, 1985) and is interpreted by many to reflect an attentional bias that favors the sampling of new locations over old (i.e., recently sampled) locations (see Klein, 2000, for a recent review).

These differences between endogenous and exogenous cuing results are consistent with the view that there are intrinsic differences between exogenous and endogenous orienting. However, the fact that these differences are primarily quantitative in nature (size and time course) has led to the more conservative interpretation that they reflect two modes of transporting a unitary attentional spotlight (Jonides, 1981; for recent reviews, see Klein, 2004; Shore & Klein, 2000). These two modes might well differ in terms of the speed and time course of the attention-orienting effect they produce, but the nature of the attention mechanism being oriented in these two modes is assumed to be the same (Jonides, 1981; see Shore & Klein, 2000, and Klein, 2004, for recent reviews).

A more powerful strategy to test whether endogenous and exogenous covert orienting are intrinsically different attentional systems looks at brain correlates of endogenous and exogenous orienting effects, with the goal of assessing whether these two kinds of orienting are mediated by similar or separate neural structures. Work with brain lesion patients and neuroimaging in nondamaged people has focused on identifying functional double dissociations, with specific brain lesions producing deficits in exogenous but not in endogenous orienting and other brain lesions producing deficits in endogenous but not exogenous orienting. Studies of patients with progressive supranuclear palsy, which is associated with lesions of the superior colliculus, have shown a selective deficit in exogenous but not in endogenous orienting (Rafal & Henik, 1994; Rafal, Posner, Friedmann, Inhoff, & Bernstein, 1988). In contrast, patients with lesions in the temporoparietal junction have shown deficits in endogenous but not in exogenous orienting (Rafal & Henik, 1994). However, more recent studies have shown that patients with lesions in the temporoparietal junction and signs of hemispatial neglect show specific deficits in exogenous orienting whereas endogenous orienting seems to be preserved (Bartolomeo & Chokron, 2002; Bartolomeo, Siéroff, Decaix, & Chokron, 2001).

A more direct comparison of brain correlates of exogenous and endogenous orienting comes from functional MRI (fMRI) studies in nondamaged people. Recent event-related fMRI studies have found very few or even no differential brain activations following endogenous and exogenous cues. In a study by Peelen and colleagues (Peelen, Heslenfeld, & Theeuwes, 2004), both cuing conditions activated the very same frontoparietal network; premotor, posterior parietal, medial frontal, and right inferior frontal cortices were equally activated for the two types of cues. This finding was interpreted as evidence in favor of a common attentional orienting mechanism. Mayer and colleagues (Mayer, Seidenberg, Dorfänger, & Rao, 2004), however, using a similar event-related fMRI procedure, found differential activations following endogenous and exogenous cues. They focused on the SOA conditions that produce maximal behavioral facilitation effects for each type of cue (i.e., short SOA conditions for exogenous cues and long SOA conditions for endogenous cues). They observed a more widespread pattern of cortical activation following endogenous than exogenous cues. At the same time, no area was exclusively activated following exogenous cues. This pattern of results was interpreted as implying that the two types of cues tap a similar attentional network but that endogenous cues recruit more areas because more effort and resources are usually required.

One limitation of these fMRI studies concerns the lack of temporal sensitivity of the blood oxygenation level dependent response. This limitation may be especially critical for the study of neural substrates of endogenous and exogenous orienting, given that the fast deployment of exogenous orienting obliges a design in which the cue and target occur within the same recorded event. Thus, as acknowledged by Mayer et al. (2004), such a situation makes it difficult to know whether brain activity differences between endogenous and exogenous cues are due to differences in the orienting mechanism per se (the generator) or to its consequences on target processing (the effect).

A third research strategy used to test the unitary attention view focuses on effects produced by exogenous and endogenous cues other than the usual facilitation effect. The logic here is that if exogenous and endogenous cues orient the very same mechanism, once it is deployed toward the target location (i.e., focusing on those levels of SOA for which facilitation effects are maximal for each type of cue), it should produce qualitatively similar effects on target processing. In contrast, qualitative differences in target processing produced by exogenous and endogenous cues would constitute evidence against a common processing view.

This research strategy has indeed revealed dissociations of a qualitative nature, with some task effects being modulated by exogenous but not endogenous cues and other effects being modulated by endogenous but not exogenous cues (see Klein, 2004; Shore & Klein, 2000, for recent reviews). Briand and Klein (Briand, 1998; Briand & Klein, 1987) found that peripheral cues produced larger cuing effects for conjunction than for feature search tasks. In contrast, central cues produced no difference in cuing effects for conjunction and feature search tasks. This result was interpreted in terms of separate attentional systems being triggered by exogenous and endogenous tasks.

A second relevant finding comes from studies focusing on the role of attention in *accelerating* stimulus processing (Carrasco & McElree, 2001). It has been argued that accelerated processing of stimuli at cued locations can make them appear to have been

presented earlier in time, as revealed by temporal order judgments (see Shore & Spence, 2005, for a recent review), and can lead to illusory percepts such as illusory line motion (Miyauchi, Shimojo, & Hikosaka, 1991; Schmidt, 2000). Recently, a dissociation between the role of endogenous and exogenous cues in producing these effects has been reported; that is, both temporal order and illusory motion effects tend to occur with exogenous cues but not with endogenous cues (Christie & Klein, 2005; K. A. Schneider & Bavelier, 2003; Shore, Spence, & Klein, 2001). The authors of these studies concluded that these temporal order and illusory motion effects are due not to endogenous attention processes but to a form of sensory facilitation triggered exclusively by peripheral spatial cues.

In contrast, other task effects have proven to be sensitive to endogenous cuing but not to exogenous cuing. Klein and colleagues have found a systematic interaction between central endogenous cuing and target frequency, with only frequent targets benefiting from attention (Handy, Green, Klein, & Mangun, 2001; Klein, 1994; Klein & Hansen, 1990). The comparison of this last dissociation with the one described earlier on visual search tasks (Briand, 1998; Briand & Klein, 1987) has recently been taken as evidence of a behavioral double dissociation (Klein, 2004; Shore & Klein, 2000).

Although the dissociations between the effects of endogenous and exogenous cues described above are difficult to reconcile with the unitary attention view, there are some weaknesses in the overall pattern of findings that are noteworthy. First, the absence of an effect for a particular type of cue could reflect a lack of sensitivity of the experimental procedure used to measure it. Second, the *double dissociation* discussed by Klein and colleagues (Klein, 2004; Shore & Klein, 2000) constitutes a dissociation between different task contexts. Task differences make it difficult to directly compare the effects produced by the two types of cues. Ideally, a dissociation would involve a single task in which performance for the same stimulus is modulated in opposite ways by exogenous and endogenous cues. To our knowledge, this type of dissociation has not been found previously.

Experiment 1

The present experiment examines the relation between exogenous and endogenous orienting in the context of a spatial Stroop task, a procedure frequently used to study the influence of irrelevant location information on performance (see Lu & Proctor, 1995, for a review). The task involves presentation of a spatial cue followed by an arrow in either the cued location or the uncued location. Subjects are required to discriminate the direction in which the peripherally presented arrow points. In our study, the arrow target appeared equally often at a congruent location (i.e., a left-pointing arrow appearing at the left side of the screen) and at an incongruent location (i.e., a left-pointing arrow appearing at the right side of the screen). For one group of participants, a peripheral noninformative cue preceded the arrow target. This type of cue was used to measure exogenous spatial orienting. For the other group of participants, a central informative cue preceded the target. The central cue consisted of one of two colored dots, with one of the colors indicating that the target was most likely to appear to the left of fixation and the other color indicating that the target was most likely to appear to the right of fixation. The color cue

correctly indicated the location of the upcoming target on 75% of the trials. The use of this type of endogenous cue was meant to prevent the contribution of automatic orienting that can occur with other symbolic cues, such as arrowheads or words indicating the most likely location (Hommel, Pratt, Colzato, & Godijn, 2001), and to maximize the likelihood of having a pure measure of endogenous spatial orienting.

If exogenous and endogenous spatial orienting have the same function, then peripheral (noninformative) and central (informative) cues should produce similar modulations of the spatial Stroop effect. Given previous findings in our laboratory (Funes & Lupiáñez, 2003; Lupiáñez & Funes, 2005), we expected exogenous cues to reduce spatial Stroop interference, so that the spatial Stroop effect ought to be smaller on valid than on invalid trials. If endogenous cues were also to reduce spatial Stroop interference, then this result would favor the unitary view. In contrast, if endogenous cues were to fail to reduce the spatial Stroop effect, or better, if they were to increase the spatial Stroop effect, then this result would be difficult to interpret in terms of the unitary view. Given the usual assumption that endogenous cues result in enhanced spatial processing at the location they cue, and given that spatial location is the interfering dimension in the spatial Stroop procedure, we thought it possible that endogenous cues would indeed increase the spatial Stroop effect. A decreased spatial Stroop predicted for exogenously cued locations together with an increased spatial Stroop effect for endogenously cued locations would constitute a dissociation between exogenous and endogenous attention.

Given the well-known differences in the time course of exogenous and endogenous orienting, four levels of cue–target SOA (100, 350, 600, and 850 ms) were used, which allowed us to examine spatial Stroop effects at the SOAs at which attentional orienting effects (either exogenous or endogenous) tend to be maximal.

Method

Participants. A total of 64 students from the University of Granada who were naive as to the purpose of the experiment participated. Their mean age was 23 years, and they all had normal or corrected-to-normal vision. They were randomly assigned to either the peripheral cue group or the central cue group.

Apparatus and stimuli. Stimuli were presented on a 15-in. (38.1-cm) color monitor connected to a PC Pentium 4. The presentation of stimuli, timing operations, and data collection were controlled by MEL2 software (W. Schneider, 1998). Participants entered a dimly illuminated room and sat at a distance of approximately 57 cm in front of the monitor. The target arrow was 11 mm in length and 13 mm in height, the boxes that marked the peripheral locations were 21 mm in height and 24 mm in width, and the inner edge of each box was 32 mm from the fixation point. The fixation point consisted of a gray circle 3 mm in diameter located at the center of the screen. In the peripheral cue group, the cue consisted of an increase in luminance of one of the boxes for 50 ms, which gave the impression of a brief flicker. In the central cue group, the cue consisted of a change in both the size and color of the fixation point. Thus, on half of the trials the fixation point became red and larger (from 3 to 7 mm in diameter), and on the other half of the trials it became green and larger.

Procedure. The sequence of events on each trial is depicted in Figure 1. At the beginning of each trial, the fixation point and the two peripheral boxes were presented for 750 ms, and then the cue appeared for 50 ms. Following the cue, the fixation point and peripheral boxes remained on the screen for 50, 300, 550, or 800 ms, depending on the SOA. The target was then presented inside one of the boxes for 33 ms. The fixation point and peripheral boxes remained alone on the screen until the participants' response, or for 1,500 ms if no response was made. Then, the next trial began. The interval between trials was 1,000 ms, and the screen remained dark gray throughout this interval. Auditory feedback was provided when an incorrect response was made.

Participants were instructed to press the *X* key on the keyboard with their left index finger when the arrow pointed to the left and to press the *M* key with their right index finger when the arrow pointed to the right. Peripheral cues appeared equally often to the left and right of fixation and were not informative about target location. In contrast, central cues were informative about the most probable target location. Thus, for half of the participants in the central cue group, a red central cue was followed by an arrow inside the right box on 75% of trials, and a green central cue was followed by an arrow target inside the left box on 75% of trials. The opposite arrangement of color information was true for the other half of the participants in the central cue group. Thus, all participants in this group received cues that were 75% predictive about target location. Participants were informed about the predictive value of the color cues and were encouraged to use this information to prepare in advance for target onset. However, in neither the peripheral cue group nor the central cue group did the cue predict the direction in which the arrow pointed; the arrow pointed equally often to the left and to the right in all conditions. All participants were instructed to respond as quickly and as accurately as possible and were strongly encouraged to maintain their eyes fixated at the center of the screen throughout the trial. The experiment was interrupted every 64 trials to allow participants a brief rest.

Design. The experiment had a mixed-factor design, with SOA, spatial congruency, and cuing manipulated within participants and

cue type manipulated between participants. The SOA variable had four levels: 100, 350, 600, and 850 ms. The cuing variable had two levels. On valid trials, the target appeared at the same location as the cue (peripheral noninformative cues) or at the location predicted by the cue (central informative cues), whereas on invalid trials the target appeared at the location opposite the cue (peripheral cues) or at the location opposite that predicted by the cue (central cues). The spatial congruency variable also had two levels. On congruent trials, the direction in which the arrow pointed was consistent with its actual location. On incongruent trials, the arrow's direction was opposite to its actual location. All levels of the SOA, spatial congruency, and cuing factors were completely crossed within blocks. Each combination of the SOA and spatial congruency factors was represented equally within each block, whereas the proportions of valid and invalid trials were set in accord with the intended predictability of cues (.50 valid for the peripheral cue group, .75 valid for the central cue group). Presentation of these trial types was mixed randomly within blocks. Participants completed 32 practice trials, followed by seven blocks of 64 experimental trials each.

Results

A mixed-design analysis of variance (ANOVA) was conducted to analyze both mean reaction times (RTs) and error percentages. The ANOVA included cuing (valid vs. invalid), SOA (100, 350, 600, and 850 ms) and spatial congruency (congruent vs. incongruent) as within-subject factors and cue type (endogenous vs. exogenous cue group) as a between-subjects factor.

Discarded data. Trials with RT faster than 100 ms (0.04% of the trials) or slower than 1,200 ms (0.40%) and trials on which an error was made (5.7%) were excluded from the RT analysis. Mean RTs were computed for each experimental condition using the remaining observations. These mean RTs and error percentages are displayed in Table 1.

RT analysis. As expected, participants were faster for valid (491 ms) than for invalid trials (512 ms), $F(1, 62) = 43.24$, $MSE = 1,230.57$, $p < .0001$, and for congruent (483 ms) than for incongruent trials (527 ms), $F(1, 62) = 193.75$, $MSE = 2,509.36$, $p < .0001$. There was also a significant main effect of SOA, with faster responses for intermediate levels of SOA (499 ms and 494 ms for the 350-ms and 600-ms SOAs, respectively) than for the 100-ms and 850-ms SOAs (517 ms and 511 ms, respectively), $F(3, 186) = 41.90$, $MSE = 687.87$, $p < .0001$.

Of interest, the $SOA \times Cuing \times Cue Type$ interaction was significant, $F(3, 186) = 8.15$, $MSE = 427.18$, $p < .0001$. Two separate ANOVAs, one for each cue type group, were conducted to examine this interaction further. The analysis for the exogenous cue group revealed a significant $Cuing \times SOA$ interaction, $F(3, 93) = 5.13$, $MSE = 345.34$, $p < .005$. The facilitation effect was largest at the shortest SOA (23 ms) and decreased linearly with increasing SOAs (11 ms, $p < .05$; 9 ms, $p < .05$; and 6 ms, $p > .10$, for the 350-, 600-, and 850-ms SOAs, respectively), $F(1, 31) = 9.63$, $p < .005$. The analysis for the endogenous cue group also revealed a significant $Cuing \times SOA$ interaction, $F(3, 93) = 3.64$, $MSE = 509.01$, $p < .05$. However, as shown in Figure 2 (left panel), this interaction was different in nature from that observed for the exogenous cue group (right panel). There was a nonsignificant facilitation effect of 5 ms at the shortest SOA ($p > .10$),

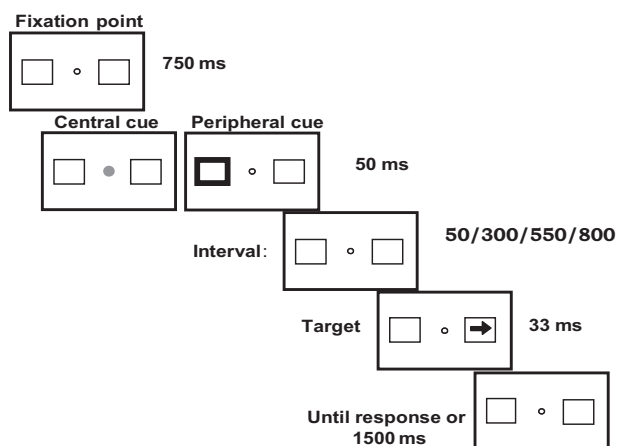


Figure 1. Schematic view of a trial sequence, from top to bottom right, for either the central cue group or the peripheral cue group, in Experiment 1. Example represents a congruent, invalid trial. The central dot in the example of central cue shown would be colored red versus green.

Table 1
Mean Reaction Times and Percentage of Errors (in Parentheses) as a Function of Congruency, SOA, Cuing, and Type of Cue in Experiment 1

SOA and congruency	Peripheral cue group		Central cue group	
	Valid	Invalid	Valid	Invalid
100 ms				
Congruent	493 (0.6%)	504 (1.0%)	492 (0.6%)	500 (1.3%)
Incongruent	513 (2.2%)	549 (3.6%)	540 (4.5%)	542 (2.4%)
350 ms				
Congruent	481 (0.3%)	489 (1.1%)	461 (0.9%)	485 (0.6%)
Incongruent	513 (2.7%)	528 (4.5%)	508 (4.7%)	527 (5.8%)
600 ms				
Congruent	473 (0.4%)	485 (0.8%)	451 (0.7%)	470 (1.5%)
Incongruent	519 (2.7%)	526 (3.9%)	507 (6.7%)	520 (7.8%)
850 ms				
Congruent	490 (0.7%)	494 (0.6%)	467 (0.6%)	498 (1.6%)
Incongruent	534 (3.1%)	543 (4.4%)	525 (7.4%)	538 (6.3%)

Note. Reaction times are in milliseconds. SOA = stimulus onset asynchrony.

which became larger and significant for the longer SOAs (22 ms, 16 ms, and 21 ms of facilitation for the 350-, 600-, and 850-ms SOAs, respectively, all p s < .005).

There was also a significant three-way interaction between cuing, spatial congruency, and cue type, $F(1, 62) = 10.94$, $MSE = 441.66$, $p < .005$, indicating that the spatial Stroop effect was modulated differently by exogenous than by endogenous cues. No other interaction reached significance. Separate three-way ANOVAs for each cue type showed that there was a significant Cuing $\times$ Spatial Congruency interaction for both the exogenous cue and the endogenous cue groups, $F(1, 31) = 4.63$, $MSE =$

447.89, $p < .05$, and $F(1, 31) = 6.38$, $MSE = 435.42$, $p < .05$, respectively. Of note, these two interactions were opposite in nature. Thus, for the exogenous cue group, there was a smaller spatial Stroop effect for valid (35 ms) than for invalid (43 ms) trials. In contrast, for the endogenous cue group, there was a larger spatial Stroop effect for valid (52 ms) than for invalid (43 ms) trials.

Further evidence for the dissociation between the effects of exogenous and endogenous cues was obtained in the analysis of RTs for the exogenous cue group. The Cuing $\times$ Spatial Congruency interaction was modulated significantly by SOA, $F(3, 93) = 3.35$, $MSE = 362.21$, $p < .05$. This interaction showed that although spatial Stroop effects were smaller for valid than for invalid trials for most levels of SOA (100, 350, and 850 ms; see Table 1 and Figure 3, right panel), this reduction in the congruency effect was most pronounced, and indeed only significant, at the 100-ms SOA (20 ms vs. 45 ms for valid and invalid trials, respectively), $F(1, 31) = 11.11$, $p < .005$. In contrast, for the endogenous cue group, the Cuing $\times$ Congruency $\times$ SOA interaction was far from significant ($F < 1$); the larger spatial Stroop effect for valid compared with invalid trials was statistically similar across all levels of SOA (see Figure 3, left panel).

Error rate analysis. The main effects of cuing and spatial congruency were similar in the accuracy analysis to those found in the RT analysis. Participants were more accurate for valid (2.4%) than for invalid trials (3%), $F(1, 62) = 5.18$, $MSE = 0.001$, $p < .05$, and for congruent (0.8%) than for incongruent trials (4.5%), $F(1, 62) = 43.72$, $MSE = 0.008$, $p < .0001$. The main effect of SOA was also significant in the analysis of errors, although the pattern was slightly different than that for RTs: Participants were most accurate at the two shortest SOAs as compared with the two longest ones, $F(3, 186) = 5.70$, $MSE = 0.001$, $p < .001$ (2%,

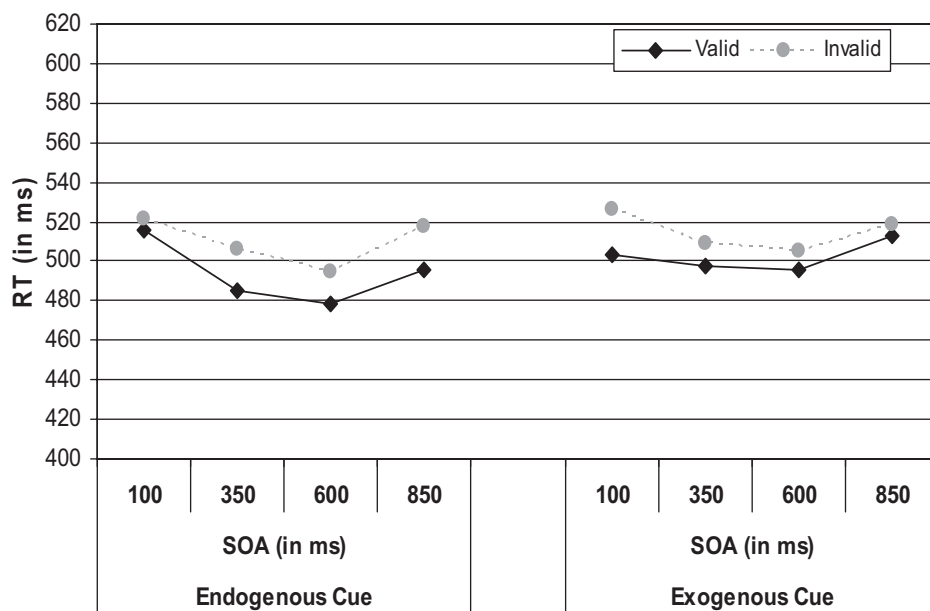


Figure 2. Mean reaction times (RTs) for valid and invalid conditions as a function of stimulus onset asynchrony (SOA), for the endogenous cue group (left) and the exogenous cue group (right) in Experiment 1.

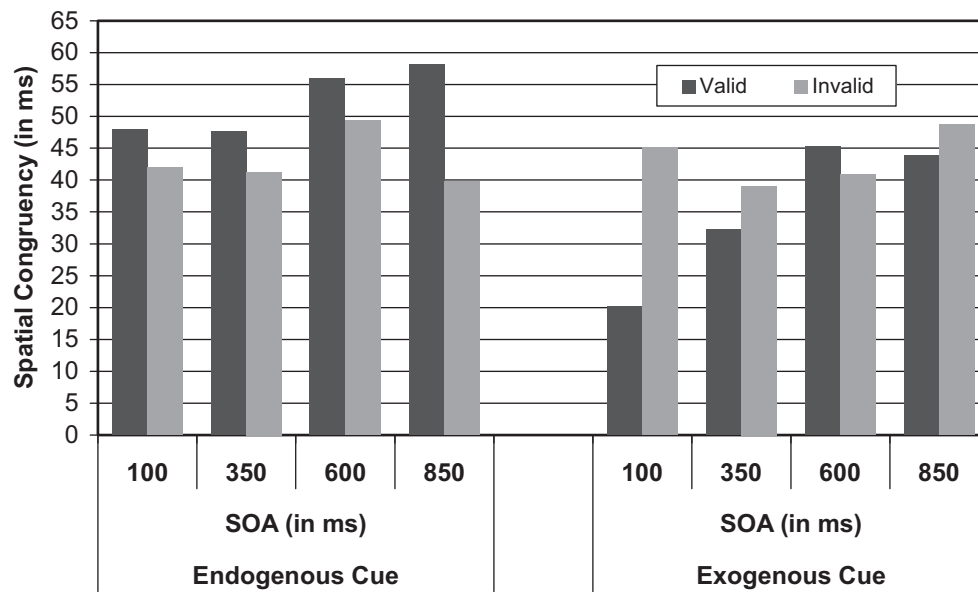


Figure 3. Spatial Stroop effect (stimulus–stimulus incongruent minus congruent trials, in milliseconds) as a function of cuing and stimulus onset asynchrony (SOA) in Experiment 1, for the endogenous cue group (left) and the exogenous cue group (right).

2.6%, 3.1%, and 3.1% for 100-, 350-, 600-, and 850-ms SOAs, respectively).

The Cuing $\times$ Spatial Congruency $\times$ Cue Type interaction did not reach significance in the accuracy analysis ($p > .10$), but it mirrored that in the RTs, with the exogenous cue group showing a smaller spatial Stroop effect for valid (2.2%) than for invalid trials (3.2%) and the endogenous cue group showing a larger spatial Stroop effect for valid (5.1%) than for invalid trials (4.3%).

Discussion

The results of the present experiment were straightforward. Regarding the time course of cuing effects, we found that it depended on the type of cue. For the exogenous cue group, a large facilitation effect was found at the shortest SOA, which decreased linearly with increasing levels of SOA. In general, this pattern of results seems consistent with the view that attention is reflexively oriented toward the cued location and rapidly reoriented, given the lack of cue predictability (Jonides, 1981; Müller & Rabbitt, 1989; Posner, Snyder, & Davidson, 1980).¹ On the other hand, for the endogenous cue group, the temporal course of cuing effects was different. For this group, a small and nonsignificant facilitation effect was observed at the shortest SOA, followed by increasing facilitation effects with increasing levels of SOA. Again, this pattern of results seems consistent with the view that an attentional orienting mechanism is voluntarily oriented toward the cued location and that this voluntary shift of attention takes more time than a reflexive shift, as endogenous cues are symbolic and must be interpreted before orienting can occur. However, once attention is deployed toward the cued location, it is sustained at that location because cues are predictive of target location (Jonides, 1981; Müller & Rabbitt, 1989; Posner et al., 1980).

Although the unitary view seems to account well for the different time courses of exogenous and endogenous cuing effects, it does not account well for the effects of exogenous and endogenous cues on the spatial Stroop effect. For the group that received peripheral noninformative cues, the spatial Stroop effects were smaller on valid than on invalid trials. In addition, the SOA manipulation allowed us to observe the time course of this Cuing $\times$ Spatial Stroop interaction. Consistent with prior studies (Lupiáñez & Funes, 2005), the reduction of the spatial Stroop effect was especially pronounced when the SOA was short and diminished as the SOA became larger (see Figure 3, right panel). In contrast, central endogenous cues modulated the spatial Stroop effect in the opposite direction; that is, there was a larger spatial Stroop effect for valid trials than for invalid trials. In addition, as is apparent in Figure 3 (left panel), the time course of the enhance-

¹ This result differed, however, from the time course of facilitation and IOR found in previous studies using other identity discrimination tasks (Lupiáñez, Milán, Tornay, Madrid, & Tudela, 1997; Lupiáñez & Milliken, 1999). In those experiments, facilitation effects were observed for longer values of SOA than in detection tasks, but they turned into IOR at SOAs larger than 500 ms. In the arrow direction discrimination task used in our study, the facilitation effect did not invert to an IOR effect even at the 850-ms SOA. It could be that this absence of IOR even at long values of SOA is due to the additional perceptual conflict present in our discrimination task. Thus, the stimulus–stimulus spatial congruency manipulation may imply an additional perceptual effort to resolve the conflict between the relevant and the irrelevant dimensions, which may delay the reorienting of attention from the cued location and consequently the appearance of IOR (Klein, 2000; see Lupiáñez et al., 2001, for a similar result using a difficult form discrimination task). Similar results were obtained by Lupiáñez and Funes (2005) with the spatial Stroop effect and by Callejas, Lupiáñez, Funes, and Tudela (2005) with the flanker effect.

ment of spatial Stroop on valid trials following endogenous cues behaves differently than the time course of the reduction of spatial Stroop on valid trials following exogenous cues. With purely endogenous cues, the amplification of spatial Stroop was similar at all levels of SOA.

The present results are difficult to reconcile with the unitary view of spatial attention and instead favor the view that exogenous and endogenous cues facilitate target processing by different means. Indeed, the results reported here constitute a *double dissociation* within the same task context. This pattern of results mitigates concerns that could be directed at the double dissociation described by Klein and colleagues, which involved different effects of endogenous and exogenous cues but across different tasks (Klein, 2004; Shore & Klein, 2000).

The larger spatial Stroop effect for endogenously valid targets seems consistent with the idea that voluntary deployment of attention amplifies the processing of location information. As spatial location constitutes the irrelevant dimension in the spatial Stroop task, it follows that the interfering effects of spatial location would be larger for valid trials, in which the distracting dimension is amplified, than for invalid trials. Further, if amplification of spatial information can be sustained voluntarily for some period of time on long cue–target SOA trials, then this would explain why the increase in spatial Stroop effects for valid trials also extends to relatively long cue–target SOAs.

The smaller spatial Stroop effect for exogenously valid targets clearly requires a very different explanation. To that end, we assume that an abrupt onset (peripheral cue) can be regarded as a perceptual object or event (Jonides & Yantis, 1988; Yantis & Jonides, 1996), sharing the spatial location with the target and contiguous in time with it. Assuming that spatial and temporal contiguity play an important role in event integration processes (Hommel, 1998; Kahneman, Treisman, & Gibbs, 1992), the facilitation effect often observed at short cue–target SOAs could be attributed, at least in part, to rapid integration of the spatial codes for the cue and the target when they occur close together in both time and space (see Funes & Lupiáñez, 2003; Funes, Lupiáñez, & Milliken, 2005; Lupiáñez & Funes, 2005; Lupiáñez & Milliken, 1999; Lupiáñez, Milliken, Solano, Weaver, & Tipper, 2001, for discussions of event integration processes in exogenous cuing contexts). Note that by interpreting exogenous cuing effects in these terms rather than exclusively in conventional “attentional orienting” terms, we can begin to explain why spatial Stroop effects might be smaller, rather than larger, for exogenously valid trials. We assume that the integration of spatial codes for cue and target within the same event file effectively separates in time the processing of the two conflicting dimensions of the target stimulus, spatial location and direction, with the distracting location dimension of the arrow target being linked with an event that occurred at an earlier point in time (the cue). The separation in time of these two perceptual codes could then underlie the reduction in the spatial congruency effect for valid trials, as the irrelevant location dimension would have largely decayed by the time the relevant direction dimension was coded (see Hommel, 1993a, for discussion of this temporal overlap hypothesis as it applies to Simon interference). Note that cue–target event integration would not occur when the cue and target appear at different locations, as would be the case for invalid trials following peripheral noninfor-

mative cues, and for both valid and invalid trials following central informative cues.

Although spatiotemporal contiguity between the cue and the target is the most obvious factor likely to constrain this integration process, we have proposed elsewhere that the contribution of such integration processes to performance is subject to strategic modulation (Funes et al., 2005; Funes, Lupiáñez, & Milliken, 2006; Funes & Lupiáñez, 2006; Lupiáñez & Milliken, 1999; Lupiáñez et al., 2001). For this reason, we would expect factors such as increases in perceptual difficulty or increases in the requirement for spatial conflict resolution to push the system toward a set that favors integration across relatively long time windows. This strategic contribution to the integration process might then explain findings of facilitation at SOAs longer than usual and trends toward spatial Stroop reduction (although rarely significant) at SOAs longer than 100 ms.

The event integration hypothesis also fits well with other effects that are observed exclusively following peripheral cues. As described in the introduction, illusory line motion effects and biased temporal order judgments are much more reliably observed with peripheral than with central cues (Christie & Klein, 2005; K. A. Schneider & Bavelier, 2003; Shore, Spence, & Klein, 2001). In the case of illusory line motion, integration of the spatial cue and the cued end of the line may result in the onset of the cued line being attributed to the onset of the cue. As a result, perception of movement from the cued end occurs (Downing & Treisman, 1997). A similar integration process may account for the cuing bias usually found on temporal order judgments. Thus, if the valid target is considered part of the cue representation, then it would be interpreted as occurring earlier, closer to the moment of onset of the cue.

In the following experiments we tested this event integration hypothesis further by introducing hybrid endogenous cues, that is, cues presented peripherally but that predicted target location. Notice that this type of cue is similar to the endogenous cues of Experiment 1 with respect to predictive value but similar to the exogenous cues of Experiment 1 with respect to spatial correspondence between cue and target. As such, these hybrid cues allowed us to tease apart two possible explanations for the results of Experiment 1. Whereas we have proposed that the different patterns of results for the two types of cues occurred because only valid exogenous cues introduced spatiotemporal correspondence between cue and target, the different patterns of results could conceivably have been caused by the differing predictive values of the two types of cues. If this were the case, then reduced spatial Stroop effects for validly cued trials should not occur with hybrid cues, which have predictive value but also introduce spatiotemporal correspondence.

Experiments 2A and 2B

In these experiments, participants again decided whether a lateralized arrow target pointed left or right, following presentation of a peripheral cue. In Experiment 2A, the peripheral cue was noninformative of the target location. In contrast, in Experiment 2B, the peripheral cue was informative of target location; on 75% of the trials, the arrow target appeared at the cued location. The purpose of these experiments was to directly compare the effects of informative and noninformative cues while holding constant

factors related to spatiotemporal correspondence. If the different results for exogenous and endogenous cues in Experiment 1 were related entirely to informativeness of the cues, then a similar pattern of results should occur here, with reduced spatial Stroop effects for valid targets when preceded by noninformative cues but larger spatial Stroop effects for valid targets when preceded by informative cues.

However, if spatiotemporal correspondence factors were key to producing the different patterns of results for exogenous and endogenous cues, then the results of the current experiments should differ from those of Experiment 1. In particular, the key comparison will be between the influence of central informative cues in Experiment 1 and the influence of peripheral informative cues in Experiment 2B. Whereas central informative cues produced larger spatial Stroop effects for valid than for invalid trials, peripheral informative cues should not necessarily produce this same result. The rationale for predicting different results across these experiments is that although the informativeness of peripheral cues in Experiment 2B could result in a larger spatial Stroop effect for valid than for invalid trials, the spatiotemporal correspondence of cue and target on valid trials should at the same time produce the opposite effect. The net result is that spatial Stroop effects could be slightly larger for valid than for invalid trials, or they could be smaller for valid than for invalid trials, this net result perhaps depending on the relative strength of two influences that push in opposite directions. These details aside, the event integration hypothesis makes one very clear prediction: The influence of peripheral informative cues on the spatial Stroop effect ought to differ from the influence of central informative cues on the spatial Stroop effect.

In addition to the predictability manipulation, we introduced a second manipulation related to the response mapping between the arrow direction and the response location. In Experiment 1, given the compatible mapping between the arrow direction and the response location (e.g., an arrow pointing left always required a left response), two sources of spatial congruency may have contributed to our measure of the spatial Stroop effect. In particular, on incongruent trials there was a stimulus–stimulus (S-S) source of spatial conflict between the irrelevant arrow location and the relevant arrow direction, as well as a stimulus–response (S-R) source of spatial conflict between the irrelevant target location and the response location. This second type of spatial congruency is usually referred to as the Simon effect (Simon & Craft, 1972; Simon, Craft, & Webster, 1973; Simon & Small, 1969; see Lu & Proctor, 1995, for a review). As such, it is unclear which of these two sources of spatial conflict was reduced by exogenous cues and increased by endogenous cues in Experiment 1.

In a recent study using peripheral noninformative cues (Luppiáñez & Funes, 2005), a compatible response mapping was directly compared with an incompatible response mapping, where participants respond with left keypresses to arrows pointing right and right keypresses to arrows pointing left. This response compatibility manipulation does not impact S-S spatial congruency, whereas it places S-R spatial congruency in opposition across the two conditions, and therefore it allowed us to determine whether the influence of spatial cues on the spatial Stroop effect was due to S-S spatial congruency, S-R spatial congruency, or both. The results of this study revealed

that peripheral noninformative cues actually reduced the S-S spatial congruency effect while leaving intact the S-R source of congruency. This finding is consistent with many studies that have shown additivity between cuing effects and the Simon effect, both with exogenous cues (Hommel, 1993b; Zimba & Brito, 1995) and with endogenous cues (Proctor, Lu, & Van Zandt, 1992; Verfaellie, Bowers, & Heilman, 1988; Wascher & Wolber, 2004; Zimba & Brito, 1995). However, studies with the type of hybrid (peripheral informative) cues used in the present experiment have produced results that are less clear. Some authors have reported the Simon effect to be insensitive to peripheral cues, even when highly predictive of the target location (Proctor et al., 1992; Verfaellie et al., 1988; Zimba & Brito, 1995), whereas other authors (Stoffer & Yakin, 1994; Umiltà & Liotti, 1987) have reported a reduction of the Simon effect on valid trials.

Given the possibility that peripheral informative and noninformative cues could modulate the spatial Stroop effect via either S-S or S-R correspondence, we introduced the compatibility manipulation described above (Luppiáñez & Funes, 2005). We expected peripheral noninformative cues to modulate S-S congruency effects independent of S-R congruency. If peripheral informative cues also modulate S-S congruency effects in this way, then this result would support the idea that informative and noninformative peripheral cues modulate spatial Stroop effects at least in part by similar means, independent of their informative value.

Method

Participants. Forty-eight students participated in these experiments (20 in Experiment 2A and 28 in Experiment 2B). The mean age was 22 for men and 21 for women. All participants were undergraduate psychology students at McMaster University. All of them had normal or corrected-to-normal vision, were naive as to the purpose of the experiment, and received credit in their psychology courses in exchange for participation.

Apparatus, stimuli, and procedure. The stimuli used in these experiments were similar to the ones used in Experiment 1, but some changes in the procedure were introduced. First, although the spatial cues used in Experiment 2A were the same as those used in the exogenous cue condition in Experiment 1 (i.e., peripheral and noninformative), the spatial cues used in Experiment 2B were peripheral and informative of the target location; on 75% of the trials, the target appeared in the same location as the cue. Second, a response compatibility manipulation was introduced. In the compatible task, participants were to press the *X* key (left response) when the arrow pointed to the left and the *M* key (right response) when the arrow pointed to the right, independent of the arrow's location. In the incompatible task, the opposite response mapping was used; that is, participants were to press the *M* key (right response) when the arrow pointed to the left and the *X* key (left response) when the arrow pointed to the right. Third, to simplify the design and reduce the length of the experiment, two rather than four levels of SOA were manipulated. The levels of the SOA variable tested in these experiments were 100 ms and 600 ms.

Design. The designs of Experiments 2A and 2B were identical, with the following exceptions. For both experiments, partici-

pants performed a block of 16 practice trials (2 trials per experimental condition) followed by a series of experimental blocks of 32 trials each (4 trials per experimental condition). In Experiment 2A, nine experimental blocks per task were completed, constituting a total of 288 trials per task (36 per experimental condition). In Experiment 2B, seven blocks per task were completed, constituting a total of 224 trials per task. In each block, there were 8 invalid trials and 24 valid trials. Thus, in total there were 14 observations in each experimental condition for invalid trials (2 per block) and 42 observations in each experimental condition for valid trials (6 per block). Experiment 2A took about 60 min to complete, and Experiment 2B took about 45 min to complete.

Results

Mean correct response latencies and error rates were computed for each participant and experimental condition. A repeated measures ANOVA was conducted for each experiment, to analyze both mean RTs and error percentages. Each ANOVA included task (compatible vs. incompatible), spatial congruency (congruent vs. incongruent), SOA (100 vs. 600 ms), and cuing (valid vs. invalid) as within-subject variables. Mean RTs and error percentages for each experimental condition are presented in Tables 2 (Experiment 2A) and 3 (Experiment 2B).

Discarded data. Trials from the practice block and those in which a miss (no response) or a wrong response was made were excluded from the RT analysis. In addition, trials with correct responses faster than 100 ms (0.44% and 0.10% for Experiments 2A and 2B, respectively) or slower than 1,200 ms (0.54% and 0.40% for Experiments 2A and 2B, respectively) were excluded from the RT analysis.

RT analysis. The ANOVA performed on the data from Experiment 2A revealed significant main effects of task, $F(1, 19) = 4.62$, $MSE = 12,527.50$, $p < .05$; spatial congruency, $F(1, 19) = 16.95$, $MSE = 2,233.13$, $p < .001$; SOA, $F(1, 19) = 7.36$, $MSE = 1,701.06$, $p < .05$; and cuing, $F(1, 19) = 8.61$, $MSE = 2,199.03$, $p < .01$. Responses were faster for the compatible (423 ms) than for the incompatible task (450 ms), for long (430 ms) than for short SOA trials (442 ms), for congruent (425 ms) than for incongruent trials (447 ms), and for valid (428 ms) than for invalid trials (444 ms). Although the Cuing $\times$ SOA interaction did not reach significance ($p > .10$), planned comparisons revealed that responses were reliably faster for valid than for invalid trials only at the short SOA (20 ms, $p < .001$; see Figure 4, left panel).

As in Experiment 1, spatial congruency interacted with cuing, $F(1, 19) = 9.93$, $MSE = 580.52$, $p < .01$, with the spatial

congruency effect being smaller on valid trials than on invalid trials (14 ms vs. 31 ms, respectively). The Spatial Congruency $\times$ Cuing $\times$ SOA interaction did not reach significance ($p > .10$). However, as shown in Figure 5 (left panel), the trend in the results was consistent with Experiment 1, with the reduction in spatial congruency effect on valid trials most pronounced at the short SOA.

Task interacted with spatial congruency, $F(1, 19) = 13.84$, $MSE = 1,939.06$, $p < .01$. The spatial congruency effect was significant for the compatible task, $F(1, 19) = 32.47$, $MSE = 1,978.44$, $p < .0001$, but not for the incompatible task ($F < 1$). More important for the aim of the experiment was the finding that the Congruency $\times$ Cuing interaction was completely independent of the task compatibility manipulation, as revealed by the null three-way interaction ($F < 1$). Thus, as can be observed in Figure 6 (left panel), the reduction in spatial congruency effect observed on valid trials did not differ for the compatible and the incompatible tasks.

In the RT analysis of Experiment 2B, there was a significant main effect of task, $F(1, 27) = 23.52$, $MSE = 11,143.83$, $p < .0001$; cuing, $F(1, 27) = 129.61$, $MSE = 2,676.76$, $p < .0001$; SOA, $F(1, 27) = 5.10$, $MSE = 2,513.21$, $p < .05$; and spatial congruency, $F(1, 27) = 32.74$, $MSE = 2,284.51$, $p < .0001$. Responses were faster for the compatible (545 ms) than for the incompatible task (593 ms), for valid (541 ms) than for invalid trials (597 ms), for long (564 ms) than for short SOA trials (574 ms), and for congruent (556 ms) than for incongruent trials (582 ms).

A null Cuing $\times$ SOA interaction ($F < 1$) indicated that facilitation was obtained at both levels of SOA. As can be observed in Figure 4 (right panel), large and equivalent facilitation effects were obtained at both levels of SOA (55 ms at both the 100- and 600-ms SOA).

Spatial congruency interacted with cuing, $F(1, 27) = 5.52$, $MSE = 859.52$, $p < .05$, with the spatial congruency effect again being smaller on valid trials than on invalid trials (19 ms vs. 32 ms, respectively). The Spatial Congruency $\times$ Cuing $\times$ SOA interaction did not reach significance ($p = .20$), but the trend was consistent with that observed in Experiments 1 and 2A, with the reduction in spatial congruency on valid trials being most obvious at the short SOA.

As in Experiment 2A, task interacted with spatial congruency, $F(1, 27) = 4.82$, $MSE = 2,138.25$, $p < .01$, indicating that the Stroop-like effect was larger with the compatible task (36 ms) than with the incompatible task (16 ms). More important,

Table 2

Mean Reaction Times and Percentage of Errors (in Parentheses) per Experimental Condition in Experiment 2A

Task	SOA 100 ms				SOA 600 ms			
	Valid		Invalid		Valid		Invalid	
	C	I	C	I	C	I	C	I
Compatible	403 (1%)	436 (6%)	413 (3%)	467 (5%)	395 (2%)	426 (6%)	401 (1%)	443 (6%)
Incompatible	452 (6%)	440 (5%)	461 (4%)	470 (6%)	439 (3%)	440 (4%)	441 (4%)	456 (6%)

Note. Reaction times are in milliseconds. SOA = stimulus onset asynchrony; C = congruent; I = incongruent.

Table 3

Mean Reaction Times and Percentage of Errors (in Parentheses) per Experimental Condition in Experiment 2B

Task	SOA 100 ms				SOA 600 ms			
	Valid		Invalid		Valid		Invalid	
	C	I	C	I	C	I	C	I
Compatible	513 (2%)	537 (5%)	564 (3%)	603 (6%)	487 (1%)	523 (4%)	544 (3%)	587 (8%)
Incompatible	565 (4%)	573 (5%)	604 (3%)	637 (6%)	561 (4%)	571 (5%)	610 (6%)	626 (8%)

Note. Reaction times are in milliseconds. SOA = stimulus onset asynchrony; C = congruent; I = incongruent.

as in Experiment 2A, the Congruency $\times$ Cuing interaction was independent of the task compatibility manipulation, as revealed by the null three-way interaction ($F < 1$) (see Figure 6, right panel).

Error rate analysis. In the analysis of errors conducted on Experiment 2A, there was a significant effect of spatial congruency, $F(1, 19) = 19.14$, $MSE = 0.002$, $p < .0005$, with participants making more errors on incongruent (5.5%) than on congruent trials (3.0%). The Task $\times$ Spatial Congruency interaction was also significant, $F(1, 19) = 5.81$, $MSE = 0.003$, $p < .05$, showing that the congruency effect was larger (and significant) in the compatible task (4%), $F(1, 19) = 18.77$, $MSE = 0.003$, $p < .0005$, than in the incompatible task (1%; $p > .10$). No other results reached significance.

The analysis of errors conducted on Experiment 2B revealed significant main effects of cuing and spatial congruency, $F(1, 27) = 9.25$, $MSE = 0.003$, $p < .01$, and $F(1, 27) = 13.41$, $MSE = 0.005$, $p < .005$, respectively. Thus, participants made fewer errors on valid (3.7%) than on invalid trials (5.3%) and on congruent (3.3%) than on incongruent trials (5.8%). In addition, the Cuing $\times$ SOA interaction was significant, $F(1, 27) = 5.60$, $MSE = 5.59$,

$p < .05$, indicating a nonsignificant facilitation effect at the 100-ms SOA (0.6%, $p > .10$), which was smaller than that observed at the 600-ms SOA (2.6%, $p < .05$). No other results reached significance.

Comparing Experiments 2A and 2B. After analyzing the results of Experiments 2A and 2B separately, we conducted an ANOVA with experiment as a between-participants variable and task compatibility, cuing, SOA, and congruency as within-participant variables. This analysis revealed that experiment interacted with cuing, $F(1, 46) = 30.51$, $MSE = 2,479.40$, $p < .0001$, such that the facilitation effect observed in Experiment 2B with predictive peripheral cues was more than three times larger than that found in Experiment 2A with noninformative peripheral cues (see Figure 4 for a visual comparison). This result indicates that the processes implicated in the facilitation effects were dependent on the predictability of the cue.

Once again, cuing interacted with spatial congruency, $F(1, 46) = 14.11$, $p < .0001$. However, this Cuing $\times$ Spatial Congruency interaction did not vary as a function of experiment, ($F < 1$; see Figure 5). This result indicates that the reduction of spatial Stroop on valid trials was independent of the predictability of the

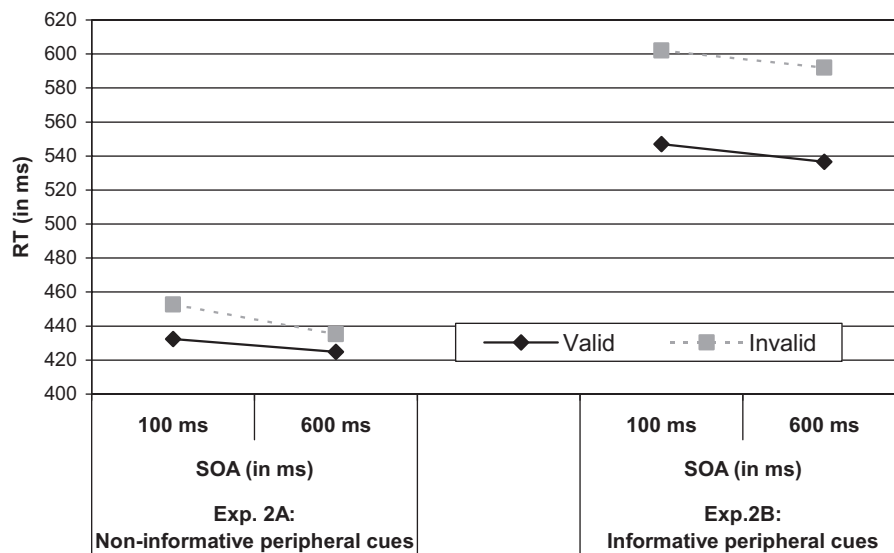


Figure 4. Mean reaction times (RTs) for valid and invalid conditions, as a function of stimulus onset asynchrony (SOA; 100 ms and 600 ms) in Experiments 2A (left) and 2B (right). Smaller RTs were found for valid than for invalid trials. Facilitation effects (faster responses for valid than for invalid trials) were found at both levels of SOA.

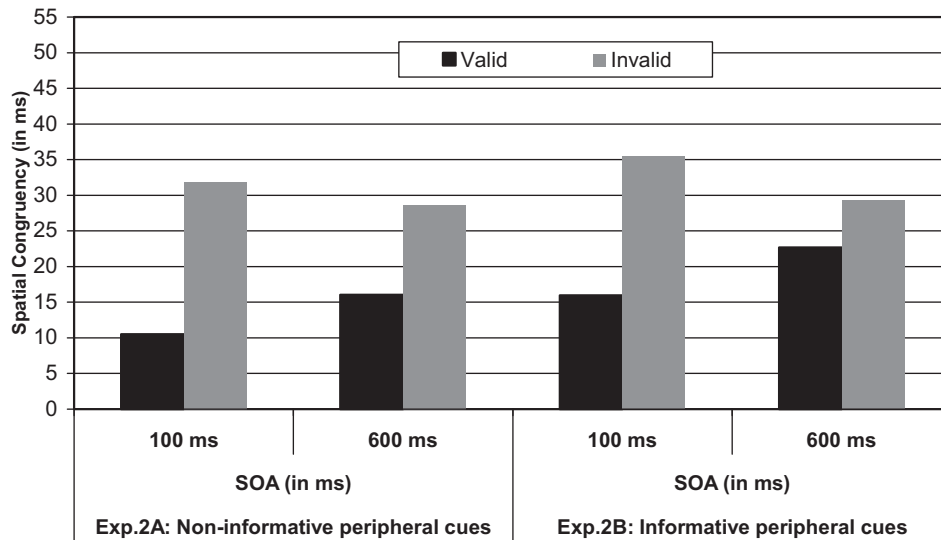


Figure 5. Spatial Stroop effect (stimulus–stimulus incongruent minus congruent trials, in milliseconds) as a function of cuing and stimulus onset asynchrony (SOA) in Experiments 2A (left) and 2B (right). Reduced spatial Stroop was found for valid compared with invalid trials in both experiments, this effect being more pronounced for the short than for the long level of SOA.

cue, a result that contrasts sharply with that observed in Experiment 1. The four-way interaction involving task (i.e., Experiment $\times$ Task $\times$ Cuing $\times$ Spatial Congruency) was clearly not significant ($F < 1$).

Discussion

The primary objective of Experiments 2A and 2B was to determine whether the opposite effects of exogenous and endogenous

cues on the spatial Stroop effect in Experiment 1 were due to differences in the informative value of the cues or to differences in cue–target spatial correspondence afforded by the cues. To address this issue, we compared the performance of two groups of participants. In Experiment 2A, participants were presented with peripheral noninformative cues, whereas in Experiment 2B, participants were presented with peripheral informative cues. These two types of cues differed in predictive value but not in terms of spatial correspondence of cue and target.

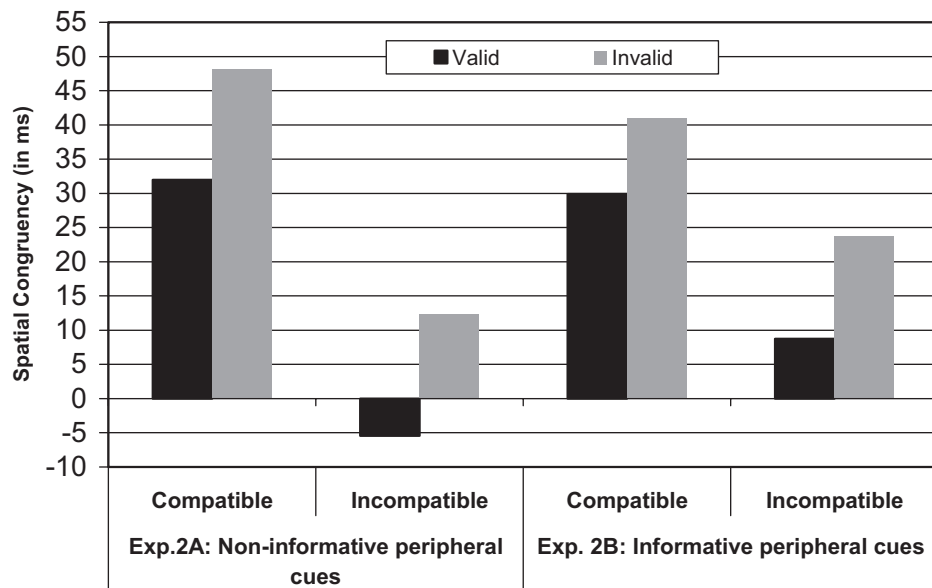


Figure 6. Spatial congruency effect as a function of cuing and response compatibility in Experiments 2A (left) and 2B (right). Reduced spatial congruency effects were observed for valid as compared with invalid trials in both experiments. This reduction was almost identical in magnitude for the compatible and incompatible tasks.

The larger facilitation effect observed for peripheral informative cues in comparison with noninformative cues confirms that the predictability of the cues did influence performance. At the same time, the key finding was that peripheral informative and noninformative cues affected spatial Stroop effects in the same manner, producing smaller spatial Stroop effects for validly cued trials. This result strongly supports the view that differences in cue–target spatiotemporal correspondence, rather than differences in cue–target predictability, were key to the opposite influences of exogenous and endogenous cues on spatial Stroop observed in Experiment 1. What is somewhat surprising is that there was no apparent influence on spatial Stroop of the predictability of the cues, whereas we had thought that perhaps spatiotemporal correspondence and predictability would produce opposing effects on spatial Stroop. This result highlights the important difference between the cuing effects produced by cues that correspond spatiotemporally with targets and cuing effects produced by cues whose correspondence depends on a symbolic relation with the target.

In contrast to what was found in Experiment 1 for the peripheral noninformative cue group, the interaction between cuing, spatial Stroop, and SOA found in Experiment 1 did not reach significance in Experiments 2A and 2B. At first blush, the absence of this interaction seems difficult to reconcile with the integration hypothesis described before, as integration is much easier to imagine for events separated by 100 ms than for events separated by 600 ms. However, the absence of this interaction in Experiments 2A and 2B does not imply that such an effect does not exist. In fact, the reduction of spatial Stroop for cued trials was numerically larger for the 100-ms SOA than for the 600-ms SOA, in agreement with the results of Experiment 1. In addition, recent data from other studies in our lab using the same paradigm have shown time and again that the reduction of spatial Stroop on peripherally cued trials tends to occur for short, 100-ms SOAs but not for longer SOAs (Funes & Lupiáñez, 2006; Lupiáñez & Funes, 2005). Taken together, the experiments in which we have studied the spatial Stroop reduction with peripheral cuing present a clear picture; the reduction in spatial Stroop is greater at the 100-ms SOA than at longer SOAs, and on most occasions this reduction is significant only at this short SOA.

Finally, a further important result found in the present experiments was that the Cuing $\times$ Spatial Congruency interaction was not altered by the response compatibility manipulation in either Experiment 2A or Experiment 2B. This result can be taken as an additional source of evidence that peripheral informative and noninformative cues modulate spatial Stroop by the same means, that is, by reducing the spatial congruency between the stimulus location and the stimulus direction (S-S). In other words, although Simon effect conflict (S-R) can contribute to our measure of the spatial Stroop effect, this component of the spatial Stroop effect appears not to play a role in the focal pattern of results discussed in this article.

General Discussion

In Experiment 1, we obtained a double dissociation between the effect of endogenous (central, symbolic, informative) and exogenous (peripheral, noninformative) cues on the spatial Stroop effect. Whereas exogenous cues reduced the spatial Stroop effect for valid

relative to invalid trials, the endogenous cues used in this experiment increased the spatial Stroop effect for valid relative to invalid trials. This double dissociation involving exogenous and endogenous cues within the same task is, in principle, difficult to reconcile with the unitary view of spatial attention.

The larger spatial Stroop effect following valid than invalid endogenous cues seems consistent with the idea that spatial attention was oriented toward the cued location and maintained at that location in accord with the predictive value of the cue. More concretely, because central symbolic cues are not presented at the potential target location, they may have led to the generation of an internal representation of the spatial location indicated by the cue, which would be maintained in working memory until the target appears. That mental representation, either verbal or visual in nature, might consist of the amplification of a code associated with the cued location, such that when congruent with the location of the arrow on validly cued trials, it would produce a particularly larger spatial Stroop effect. A number of studies have found that creating a memory representation of a specific spatial location (but not when memorizing any other feature at that location) results in benefits in processing for the stimuli appearing at that location similar to the ones produced by endogenous attention (Awh & Jonides, 2001). Electrophysiological and neuroimaging studies have also found evidence linking spatial working memory with effects produced by informative, symbolic cues (Corbetta, Kincade, & Shulman, 2002; Nobre et al., 2004). A related form of mental representation that could be triggered by symbolic cues is a directional “left–right” code from the center toward the location indicated by the cue, which would work as an additional source for congruency on valid trials. In this case, the direction of the target arrow could then be congruent or incongruent not only with respect to its actual location but also with respect to the directional representation created by the symbolic cue. The creation of this additional directional code could therefore account for the larger spatial Stroop effects on cued trials following central informative cues. We propose that this spatial/directional representation triggered by central endogenous cues may constitute the essence of the endogenous orienting process itself, a process that in the present study was elicited exclusively by central/symbolic, informative cues.

In contrast, it follows that the creation of this type of directional code would not be elicited by peripheral cues, as these cues appear at the peripheral target location itself. Indeed, the fact that central and peripheral cues in Experiment 1 affected spatial Stroop in opposite ways is consistent with the idea that central and peripheral cues produce their effects through different types of internal representations. We propose that the spatiotemporal correspondence between cue and target on valid exogenously cued trials results in integration of the perceptual spatial codes for the cue and the target. This integration process *speeds* processing of the spatial location of the target, which in turn reduces the temporal overlap in processing of the target’s spatial location and direction. The separation in time of these two perceptual codes could then underlie the reduction in the spatial congruency effect for valid trials.

The results of Experiments 2A and 2B were consistent with the view that spatiotemporal correspondence between cue and target played a key role in the exogenous cuing results observed in Experiment 1. In both of these experiments the cues were peripheral abrupt onsets, but these cues were informative of target

location only in Experiment 2B. The spatial Stroop effect was smaller for valid than for invalid trials in both experiments, suggesting that spatial correspondence between the cue and target was critical for the reduction of spatial Stroop on valid exogenously cued trials in Experiment 1. More important, the opposite effects on spatial Stroop produced by peripheral informative cues in Experiment 2B and central informative cues in Experiment 1 suggest that although these two types of cues (peripheral and central) were equally predictive, they facilitated target processing by different means. Again, the view that central and peripheral informative cues produce their effects by virtue of different spatial representations seems applicable here. More concretely, for central informative cues (Experiment 1), the critical spatial representation may be endogenously generated, producing both faster RTs for valid than for invalid trials and larger spatial Stroop effects for valid than for invalid trials. In contrast, for peripheral informative cues (Experiment 2B), a spatial expectancy may be implemented without endogenously generating an internal representation in response to the cue but relying instead on the same perceptual representation that underlies performance following peripheral noninformative cues (Experiments 1 and 2A). The use of a common perceptual representation for peripheral informative and noninformative cues would explain why the reduction in size of the spatial Stroop effect for valid trials was equivalent in Experiments 2A and 2B.

The final result that requires an explanation is why the facilitation effect is larger for peripheral informative than for peripheral noninformative cues, whereas the reduction in the spatial Stroop effect for valid trials was similar for peripheral informative and peripheral noninformative cues. We speculate that that predictiveness of the cue, or perhaps simply the greater frequency of valid relative to invalid trials, can alter the size of cuing effects while having very little effect on the temporal overlap between the cue and target spatial representations. As it is the latter of these processes that we propose is responsible for the reduction of spatial Stroop at the location of the peripheral cue, modulations of spatial Stroop need not necessarily be perfectly correlated with modulations of peripheral cuing effects.

Does the argument described above to explain the qualitatively opposite influences of central and peripheral cues on the spatial Stroop effect found in this study completely rule out the unitary view? Our claim is that the present results constitute evidence against the view that the same attentional mechanism is oriented by endogenous and exogenous cues, insofar as these two types of cues appear to make use of different spatial representations. Indeed, it is hard to imagine how the orienting of one and the same mechanism could produce opposite influences on spatial Stroop. However, if the unitary view were to allow that endogenous and exogenous cues trigger processes apart from the orienting of attention, then this view could accommodate the present results. For example, it could be argued that central and peripheral cues orient one and the same attentional mechanism but that this mechanism is not the one responsible for modulation of the spatial Stroop effect. In addition to the orienting process, which might arguably be common to central and peripheral cues, other processes would then be invoked to explain why the spatial Stroop effect is sensitive to cuing manipulations. For example, it could be argued that the event integration process proposed above is elicited exclusively by peripheral cues independent of their predictive

value and that this integration process is the key source of the reduction of spatial Stroop on valid trials. However, this process could be triggered by peripheral cues independent of the orienting of attention. A similar argument might be forwarded regarding dual influences of central symbolic cues on performance. That is, in addition to eliciting shifts of attention, central informative cues could trigger other processes (e.g., activation of an internal directional code) that are responsible for increasing the spatial Stroop effect on validly cued trials but that do not contribute to cuing effects.

Regardless of which of these theoretical accounts is more accurate, it seems clear that spatial Stroop effects provide a window into the distinct processes elicited by central and peripheral cues. The double dissociation observed here between the influences of these two types of cues on spatial Stroop has important implications for how we interpret the results of spatial cuing studies. In particular, the present results force us to be cautious about attributing influences of a spatial cue simply to the orienting–reorienting dynamics of attention. The present results show clearly that central and peripheral cues can affect performance through very different processes.

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